Ninad Alurkar

+1 (248) 990-5119 | ninadalurkar4@gmail.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Senior Robotics Software Engineer with **5+ years of experience** specializing in autonomous systems, quadruped locomotion, and real-time robotics software development. Experienced in building **ROS2-based** autonomy stacks using **C++** and **Python**, with strong expertise in **SLAM**, **sensor fusion**, **motion planning**, and embedded control systems. Skilled in developing perception and state estimation pipelines using **EKF**, **factor graph optimization**, and **computer vision (OpenCV, PyTorch)**, along with deploying optimized robotics solutions on edge platforms using **CUDA**, **TensorRT**, and containerized **Docker-based CI/CD** workflows.

TECHNICAL SKILLS

Robotics & Autonomy: ROS2 (Nav2, Lifecycle Nodes, DDS/FastDDS), MoveIt, OMPL, Behavior Trees (BT.CPP), SLAM (LiDAR/Visual), Trajectory Optimization (MPC, RRT*, CHOMP), Multi-Robot Systems
Perception & AI: Computer Vision (OpenCV), Deep Learning (PyTorch, TensorFlow), Object Detection (YOLO-style models), Visual SLAM, 3D Mapping, TensorRT, ONNX Runtime
Control & Embedded Systems: C/C++, Python, Real-time Control Systems, Sensor Fusion (EKF/UKF), ARM Cortex, FPGA Integration, FreeRTOS, Motor Control Systems
Simulation & Robotics Testing: Gazebo, Isaac Sim, NVIDIA Omniverse, HIL/SIL Testing, Digital Twins
Infrastructure & Tools: Docker, Kubernetes (light), Git, CI/CD (GitLab/GitHub Actions), CUDA, MATLAB, Agile Robotics Development, Edge Deployment Systems

PROFESSIONAL EXPERIENCE

Robotics Software Engineer, Boston Dynamics. **Sep 2025 – Present | United States**

- Led **quadruped locomotion control optimization**, developing real-time control loops using **ROS2 (DDS/FastDDS)** and **C++**, reducing control-loop jitter by **26%** through deterministic scheduling, real-time threading, and executor optimization.
- Designed **multi-sensor state estimation system** integrating **IMU, force-torque sensors, and stereo vision**, using **EKF and factor graph optimization**, improving pose stability accuracy by **21%** in high-dynamic terrain traversal.
- Developed motion planning and trajectory optimization stack using **OMPL** and **MPC-based controllers** for reliable navigation in complex environments with strict real-time constraints.
- Built **real-time perception pipeline** for terrain segmentation and obstacle detection using **OpenCV and PyTorch**, improving detection robustness by **23%** in low-light and motion-blurred environments.
- Implemented **simulation-to-real validation framework** using **Gazebo and Isaac Sim**, increasing edge-case scenario coverage by **2.8x** and significantly reducing physical robot testing iterations.
- Programmed edge autonomy deployment stack using containerized **Docker-based ROS2** systems with **TensorRT-optimized inference pipelines** for real-time perception and control on embedded robotic platforms.
- Contributed to **system-level robotics autonomy architecture**, integrating **behavior trees (BT.CPP), ROS2 lifecycle nodes, and modular navigation components**, improving runtime reliability and system determinism across autonomy pipelines.

Robotics Software Engineer, Honeywell **Jan 2021– Jul 2024 | India**

- Architected **ROS2-based autonomous navigation system** using **DDS middleware and modular node design**, reducing motion planning latency by **38%** in warehouse robotics applications.
- Constructed **multi-modal SLAM pipeline** integrating **LiDAR, stereo vision, and IMU sensors**, using **EKF and graph-based optimization**, improving localization accuracy by **27%** in dynamic industrial environments.
- Established **CI/CD pipeline for robotics systems** using **Docker and GitLab CI**, with automated **Gazebo simulation testing**, reducing deployment cycle time by **42%**.
- Applied **sensor fusion algorithms (EKF/UKF)** for real-time pose estimation, reducing drift error (RMSE) by **19%** during long-duration autonomous operations.
- Optimized **edge AI perception pipeline** using **CUDA, TensorRT, and PyTorch model compression**, reducing inference latency by **31%** on embedded robotics hardware.
- Engineered motion planning system using **MoveIt**, **OMPL**, and **ROS2 Control** for reliable trajectory execution in constrained multi-robot environments.
- Developed **Gazebo-based simulation environment** for navigation and manipulation testing, improving scenario coverage and strengthening pre-deployment validation for robotic systems
- Executed cloud robotics telemetry system **using** MQTT-based streaming architecture**, enabling real-time monitoring across** 120+ deployed robotic systems**.**

EDUCATION

Master of Science in Robotics Engineering | **University of Michigan – Dearborn, United States**
Bachelor of Technology in Mechanical Engineering | **National Institute of Technology – Surat, India**